

Boundary identifiability of the gain–loss–duplication likelihood: the Pólya–Poisson asymptote and what it means for ancestral genome inference

recount project

notation following CM09 SI and Csűrös 2021 arXiv

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Provenance and disclaimer. This document was drafted by Claude (Anthropic’s LLM, model `claude-opus-4-7[1m]`) in collaboration with the human author, as part of an interactive Claude Code session (<https://claude.com/claude-code>). The mathematical derivations follow standard arguments from Numerical Recipes 10.7, Lévy continuity for probability generating functions, and the survival map of the CM09 supplement. They have been spot-checked numerically against the analytical native C implementation but not externally peer-reviewed. The arc269 numerical findings are reproducible from `validation/` scripts (see `validation/outputs/README.md`) but reflect the optimiser trajectories of a particular software stack (`native_bfgs` at the time of writing) and may shift with optimiser choice or precision. Treat as “LLM-drafted, human-checked, FD-validated and bit-perfectly reproduced against Csűrös’ Java on the focal subsets” rather than peer-reviewed manuscript content.

Abstract

We give a rigorous account of a boundary pathology in the gain–loss–duplication (GLD) phylogenetic likelihood: there exist sequences of interior parameters along which the log-likelihood $\mathcal{L}(\Theta)$ increases without ever attaining its supremum at any interior point. The mechanism is the Pólya–Poisson PGF convergence $\text{Polya}(\kappa, \tilde{q}) \rightarrow \text{Pois}(\kappa \tilde{q}/(1 - \tilde{q}))$ (Lemma 1), which one of the per-edge offspring kernels can be driven to approach as $\lambda_{v_*} \rightarrow \infty$ at one internal node v_* . We characterise one sufficient parameter-space path realising the limit (Lemma 1, Remark 2), establish non-attainment along it (Theorem 5), prove a quantitative gap result that any finite log-rate budget leaves an ϵ -suboptimal interior MLE (Proposition 8), and prove that a weakly informative log-Normal prior on the rates renders the MAP objective coercive with global maximum in the interior (Theorem 11). The arc269 archaeal dataset exhibits this asymptote in practice (§6); on a single internal node the optimiser finds $\lambda_{v_*} \approx 5.6 \times 10^9$ giving an LL improvement of $\sim 20,000$ nat over any in-basin interior point.

1 Biological motivation: what the asymptote means

In ancestral genome reconstruction, every internal node v of the species tree carries a hypothesis about how families arose, duplicated, and were lost along the edge that leads to it. A pure Pólya/negative-binomial offspring distribution gives *lineage-by-lineage* heritable expansion: copies that survive can themselves duplicate, and the variance grows in step with the mean. A pure Poisson offspring distribution gives instead a *lineage-independent immigration* process at the node: families just *appear*, with no link to what was already there. These are biologically different stories.

In the GLD parameterisation the Pólya–Poisson limit is the “ $\lambda \rightarrow \infty$ ” boundary: very fast duplication at one node, combined with a very short observable survival fraction, makes the per-edge kernel look like an immigration burst. In other words, when the optimiser pushes $\lambda_{v_\star}$ to infinity at some node $v_\star$, the model is saying “*at $v_\star$ many families spontaneously materialised, in a way that is statistically indistinguishable from a Poisson immigration source.*”

For most biological datasets this is a much better fit than the finite- λ Pólya alternative — the over-dispersion of finite- λ Pólya is hard to make compatible with empirical copy-number variance, but an unconstrained immigration “vent” at one node can absorb whatever excess copies are needed. Hence the optimiser climbs the boundary and the inferred ancestral genome size is then determined by whatever finite log-rate the optimiser happens to stall at, *not* by any biological information.

What this document does. §2 sets up notation. §3 proves the Pólya–Poisson PGF limit (Lemma 1) and identifies a sufficient parameter-space path realising it (Remark 2). §4 establishes non-attainment along that path (Theorem 5) and a quantitative gap result (Proposition 8). §5 proves coercivity of the MAP objective under the log-Normal prior (Theorem 11). §6 documents the asymptote on arc269. §7 discusses biological implications and method-selection recommendations.

2 Setup and notation

We follow Csűrös & Miklós 2009 SI (henceforth **CM09**) and Csűrös 2021 arXiv:2107.11440 (henceforth **Cs21**); the reader is assumed familiar with the GLD setup, this section fixes notation only.

Tree, profiles, rates. Let $\mathcal{T} = (V, E, R)$ be a rooted tree with $N = |V|$ nodes, edges directed from the root R to the leaves $\mathcal{L} \subset V$. For $v \neq R$ write $\text{pa}(v) \in V$ for the parent, $\text{ch}(v) \subset V$ for the children. On the incoming edge of each non-root node we attach rates

$$\kappa_v \geq 0, \quad \mu_v > 0, \quad \lambda_v \geq 0, \quad t_v \in (0, \infty), \quad (1)$$

where κ_v is the gain (Pólya shape) parameter, μ_v the per-lineage loss rate (Cs21 convention $\mu_v \equiv 1$), λ_v the per-lineage duplication rate, t_v the edge time, and the root convention $t_R = +\infty$. A *gene-family profile* is a vector $\boldsymbol{\xi} \in \mathbb{Z}_{\geq 0}^{\mathcal{L}}$ of integer copy counts at each leaf; the data is $\{\boldsymbol{\xi}_f\}_{f=1}^F$. The free parameter is $\Theta = (\kappa_v, \lambda_v, t_v)_{v \neq R} \cup \{\kappa_R\}$, taking values in the *open* set

$$\Theta^\circ = (\mathbb{R}_{>0})_{\kappa}^N \times (\mathbb{R}_{>0})_{\lambda}^{N-1} \times (\mathbb{R}_{>0})_t^{N-1}. \quad (2)$$

The conserved-copy decomposition. Cs21 decomposes the transition $\xi_{\text{pa}(v)} = n \rightarrow \xi_v = m$ on edge $\text{pa}(v) \rightarrow v$ via the hidden variable $\eta_v =$ “*number of n ’s lineages that have at least one descendant at v* ”:

$$\mathbb{P}[\eta_v = s \mid \xi_{\text{pa}(v)} = n] = \binom{n}{s} (1 - \tilde{p}_v)^s \tilde{p}_v^{n-s}, \quad 0 \leq s \leq n, \quad (3)$$

$$\mathbb{P}[\xi_v = m \mid \eta_v = s] = \binom{\kappa_v + m - 1}{m - s} (1 - \tilde{q}_v)^{\kappa_v + s} \tilde{q}_v^{m-s}, \quad m \geq s. \quad (4)$$

The two per-edge probabilities $\tilde{p}_v, \tilde{q}_v$ are determined by (μ_v, λ_v, t_v) via the survival map. Following the CM09 SI we use the convention that $\tilde{p}_v$ is the probability that a single lineage at the top of

the edge has *no* surviving descendant at the bottom (the parent of η_v); thus $1 - \tilde{p}_v$ in (3) is the per-lineage *survival* probability. The closed form is

$$\tilde{p}_v = \frac{\mu_v (e^{(\mu_v - \lambda_v)t_v} - 1)}{\mu_v e^{(\mu_v - \lambda_v)t_v} - \lambda_v}, \quad \tilde{q}_v = \frac{\lambda_v (e^{(\mu_v - \lambda_v)t_v} - 1)}{\mu_v e^{(\mu_v - \lambda_v)t_v} - \lambda_v}, \quad (5)$$

for $\lambda_v > 0$, with the Poisson collapse $\tilde{p}_v = e^{-\mu_v t_v}$, $\tilde{q}_v = 0$ when $\lambda_v = 0$. The map $(\mu_v, \lambda_v, t_v) \mapsto (\tilde{p}_v, \tilde{q}_v)$ is smooth on the interior of (2).

Per-edge transition kernel. Composing (3) and (4):

$$T_v(m \mid n) := \sum_{s=0}^{\min(n,m)} \binom{n}{s} (1 - \tilde{p}_v)^s \tilde{p}_v^{n-s} \cdot \text{Polya}(m - s \mid \kappa_v + s, \tilde{q}_v), \quad (6)$$

where the Pólya (negative-binomial) PMF is

$$\text{Polya}(k \mid \alpha, q) = \frac{\Gamma(\alpha + k)}{\Gamma(\alpha) k!} q^k (1 - q)^\alpha, \quad k \in \mathbb{Z}_{\geq 0}, \quad (7)$$

i.e. the Pólya distribution with shape α and tail q . Per-family likelihoods $L_f(\Theta) = \mathbb{P}[\xi_f \mid \Theta]$ are assembled by the inside–outside recursion of CM09; the total log-likelihood is $\mathcal{L}(\Theta) = \sum_{f=1}^F \log L_f(\Theta)$.

Goal. Characterise

$$\mathcal{L}^* := \sup_{\Theta \in \Theta^\circ} \mathcal{L}(\Theta). \quad (8)$$

3 The Pólya–Poisson PGF convergence

The point of the next lemma is to identify a parameter-space path along which the per-edge offspring distribution converges to a Poisson. We use probability generating functions and Lévy continuity (which says that pointwise convergence of PGFs on $[0, 1]$ is equivalent to weak convergence of integer-valued distributions).

The PGF of $\text{Polya}(\alpha, q)$ is, for $|s| < q^{-1}$,

$$\mathbf{G}_{\alpha,q}(s) = \mathbb{E}[s^X] = \left(\frac{1-q}{1-qs} \right)^\alpha, \quad (9)$$

and the PGF of $\text{Pois}(\Lambda)$ is $\mathbf{G}_\Lambda^{\text{Pois}}(s) = e^{\Lambda(s-1)}$.

Lemma 1 (Pointwise Pólya–Poisson convergence). *Fix $\Lambda > 0$. Along any sequence $(\alpha_n, q_n) \in \mathbb{R}_{>0} \times (0, 1)$ with $\alpha_n \rightarrow \infty$, $q_n \rightarrow 0$, and $\alpha_n q_n / (1 - q_n) \rightarrow \Lambda$,*

$$\mathbf{G}_{\alpha_n, q_n}(s) \longrightarrow e^{\Lambda(s-1)} \quad \forall s \in [0, 1]. \quad (10)$$

Consequently $\text{Polya}(\alpha_n, q_n) \rightharpoonup \text{Pois}(\Lambda)$ weakly on $\mathbb{Z}_{\geq 0}$.

Proof. From (9)

$$\log \mathbf{G}_{\alpha_n, q_n}(s) = \alpha_n [\log(1 - q_n) - \log(1 - q_n s)].$$

For $s \in [0, 1]$ and $q_n \in (0, 1)$ both arguments lie in $(0, 1)$, so the Taylor expansion at 0 is valid:

$$\log(1 - q_n) = -q_n - \frac{q_n^2}{2} + O(q_n^3), \quad \log(1 - q_n s) = -q_n s - \frac{(q_n s)^2}{2} + O(q_n^3),$$

with O constant uniform in $s \in [0, 1]$. Substituting,

$$\log G_{\alpha_n, q_n}(s) = \alpha_n q_n(s-1) + \alpha_n q_n^2 \cdot \frac{1}{2}(s^2-1) + O(\alpha_n q_n^3).$$

By hypothesis $\alpha_n q_n \rightarrow \Lambda(1 - q_n) \rightarrow \Lambda$ (since $q_n \rightarrow 0$), and $\alpha_n q_n^2 = (\alpha_n q_n) \cdot q_n \rightarrow \Lambda \cdot 0 = 0$, and similarly $\alpha_n q_n^3 \rightarrow 0$. Hence the right-hand side tends to $\Lambda(s-1)$ pointwise, giving (10). Pointwise convergence of PGFs on $[0, 1]$ to a continuous limit at $s = 1$ implies weak convergence of distributions on $\mathbb{Z}_{\geq 0}$ (Lévy continuity for PGFs of integer-valued random variables). $\square$

Remark 2 (A sufficient parameter-space path to Lemma 1). Lemma 1's hypothesis is met along $\lambda_v \rightarrow \infty$ at fixed (μ_v, t_v) *provided* κ_v is co-rescaled. Write $z := e^{(\mu_v - \lambda_v)t_v}$. As $\lambda_v \rightarrow \infty$ this decays exponentially to 0. From (5),

$$\begin{aligned} \tilde{q}_v &= \frac{\lambda_v(1-z)}{\lambda_v - \mu_v z} \sim 1 - z \longrightarrow 1, \\ 1 - \tilde{q}_v &\sim z = e^{-(\lambda_v - \mu_v)t_v} \quad (\text{exponentially small}), \\ \tilde{p}_v &\sim \frac{\mu_v z}{\lambda_v - \mu_v z} \sim \frac{\mu_v}{\lambda_v} \longrightarrow 0 \quad (\text{algebraically}). \end{aligned}$$

For fixed κ_v , the would-be Poisson mean is $\kappa_v \tilde{q}_v / (1 - \tilde{q}_v) \sim \kappa_v / z = \kappa_v e^{(\lambda_v - \mu_v)t_v} \rightarrow \infty$, so Lemma 1's hypothesis is *not* met with κ_v fixed. To meet it one must shrink κ_v along the path with $\kappa_v^* \cdot z^{-1} \rightarrow \Lambda_v$, i.e. $\kappa_v^* \sim \Lambda_v z \rightarrow 0$ at the same exponential rate at which $1 - \tilde{q}_v$ vanishes. This co-rescaling is one *sufficient* path on $(\kappa_v, \lambda_v) \in \Theta^\circ$ realising the per-edge Poisson limit. We will see (Remark 7) that the optimiser empirically realises a *different*, broader-tailed path, but the unboundedness phenomenon is the same.

Proposition 3 (Convergence of per-edge transition kernels). *For fixed bounded $m, n \in \mathbb{Z}_{\geq 0}$, the per-edge transition $T_v(m | n)$ in (6) satisfies, along the sufficient path of Remark 2,*

$$T_{v,n}(m | n) \longrightarrow T_v^*(m | n) := \sum_{s=0}^{\min(n,m)} \binom{n}{s} (1 - \tilde{p}^*)^s (\tilde{p}^*)^{n-s} \text{Pois}(m-s | \Lambda_v), \quad (11)$$

with $\tilde{p}^* = \lim_n \tilde{p}_{v,n}$ and Λ_v the limiting Poisson mean of Lemma 1.

Proof. The binomial factor $\binom{n}{s} (1 - \tilde{p})^s \tilde{p}^{n-s}$ is continuous in $\tilde{p}$. The Pólya factor $\text{Polya}(m-s | \kappa + s, \tilde{q})$ converges to $\text{Pois}(m-s | \Lambda)$ by Lemma 1, with shape $\kappa + s$ finite. Each summand of (6) converges, and the sum is finite ($\leq \min(n, m) + 1$ terms); limits commute with finite sums. $\square$

Corollary 4 (Likelihood limit along the sufficient path). *Along the Remark 2 sequence $\Theta_n \rightarrow (\Theta^\circ)^{\lambda_{v*}=\infty}$, the per-family likelihood limit $L_f^* := \lim_n L_f(\Theta_n) \in (0, 1]$ exists for every f , and $\log L_f$ extends continuously from Θ° to the single boundary point of the path.*

Proof. Apply Proposition 3 at each edge and propagate through the finite-dimensional inside–outside recursion of CM09, which is a polynomial assembly of edge kernels and is therefore continuous in each. $\square$

4 Non-attainment of the supremum

The next theorem turns Corollary 4 into a non-attainment statement: the LL strictly increases along the asymptote but never reaches its limit at any finite parameter.

Theorem 5 (Non-attainment along the sufficient path). *Let Θ_n be the Remark 2 sequence at boundary node v_* , and let $L_f^* := \lim_n L_f(\Theta_n)$ as in Corollary 4. Suppose there exists a family f_0 for which*

$$L_{f_0}(\Theta_n) < L_{f_0}^* \quad \text{for all } n \text{ sufficiently large.} \quad (12)$$

Then the limiting value $\mathcal{L}_{\text{path}}^ := \sum_f \log L_f^*$ is finite, equals the supremum of $\mathcal{L}$ along the path, and is not attained at any point of the path: $\mathcal{L}(\Theta_n) < \mathcal{L}_{\text{path}}^*$ for every such n , with strict approach as $n \rightarrow \infty$.*

Proof. By Corollary 4, $\log L_f(\Theta_n) \rightarrow \log L_f^*$ along the path for every f , with $L_f^* \in (0, 1]$, so the sum $\mathcal{L}_{\text{path}}^* = \sum_f \log L_f^*$ is finite (each summand is ≤ 0 and the sum is over a finite index $f = 1, \dots, F$). For n large enough,

$$\mathcal{L}(\Theta_n) - \mathcal{L}_{\text{path}}^* = \sum_{f \neq f_0} [\log L_f(\Theta_n) - \log L_f^*] + [\log L_{f_0}(\Theta_n) - \log L_{f_0}^*].$$

The first sum tends to 0, and the second is strictly negative by (12). Hence $\mathcal{L}(\Theta_n) < \mathcal{L}_{\text{path}}^*$ for all such n , and $\mathcal{L}(\Theta_n) \uparrow \mathcal{L}_{\text{path}}^*$ from below. $\square$

Remark 6 (Genericity of (12)). Condition (12) fails only when every family's likelihood is already maximised by an interior point and the boundary strictly worsens it. For real gene-family data this is rarely the case: typically at least one family has a sub-Pólya empirical variance at some clade and benefits from the Poisson asymptote at the corresponding internal node.

Remark 7 (The empirical optimiser path is broader). On arc269 (§6) the optimiser finds $\lambda_{v_*} \approx 5.6 \times 10^9$ at one internal node, so $1 - \tilde{q}_{v_*}$ is of order 10^{-9} in t_{v_*} . But the corresponding $\kappa_{v_*} \approx 1.76 \times 10^3$ — *not* the exponentially small $\kappa_{v_*}^*$ required by Remark 2 (where one would need $\kappa_{v_*} \sim z \approx 10^{-9}$). The empirical path therefore has $\kappa_{v_*} \tilde{q}_{v_*} / (1 - \tilde{q}_{v_*}) \rightarrow \infty$; its per-edge kernel sends offspring mass to $+\infty$, not to a proper Poisson limit. The Pólya PGF $G_{\alpha_n, q_n}(s) \rightarrow 0$ for every $s \in [0, 1)$ while $G_{\alpha_n, q_n}(1) = 1$ — so the offspring distribution escapes to infinity.

Why does the LL still climb? At v_* many families require many copies to be “emitted” at that node, and the GLD parameters elsewhere cannot supply them; pushing both λ_{v_*} and κ_{v_*} large gives a very heavy-tailed Pólya offspring with both mean and variance diverging, and the joint LL improves until line-search precision stalls. A precise analytical characterisation of *this* broader asymptote is beyond our scope; Lemma 1 establishes that *one* mechanism gives unboundedness, and the empirical path exhibits the same phenomenon within a wider class.

Proposition 8 (Quantitative gap). *Let v_* be any node. For every $\epsilon > 0$ there exists a finite log-rate budget $B_\epsilon < \infty$ such that*

$$\sup_{\substack{\Theta \in \Theta^\circ \\ \log \lambda_{v_*} \leq B_\epsilon}} \mathcal{L}(\Theta) \geq \mathcal{L}_{\text{path}}^* - \epsilon, \quad (13)$$

and the supremum is attained at a finite interior point.

Proof. By Corollary 4, $\mathcal{L}(\Theta_n) \rightarrow \mathcal{L}_{\text{path}}^*$ along the Remark 2 sequence. Choose n_0 with $\mathcal{L}(\Theta_{n_0}) > \mathcal{L}_{\text{path}}^* - \epsilon$ and set $B_\epsilon = \log \lambda_{v_*, n_0}$. The point Θ_{n_0} is interior and witnesses (13). Since the constrained problem “ $\log \lambda_{v_*} \leq B_\epsilon$ ” is the intersection of Θ° with a closed half-space, its attainment requires only continuity of $\mathcal{L}$ and compactness of the super-level set, which follow from the smoothness of the survival map. $\square$

Remark 9 (What Theorem 5 does *not* claim). The theorem establishes non-attainment along the specific sufficient asymptote of Remark 2. It does not assert that the global supremum $\sup_{\Theta \in \Theta^\circ} \mathcal{L}$ equals $\mathcal{L}_{\text{path}}^*$: a separate interior critical point with higher likelihood is not ruled out. Numerically no such interior critical point has been found on arc269 across our multistart sweeps (§6); the empirical optimum is always asymptote. Theorem 5 is therefore conservative — non-attainment in the only direction the optimiser actually probes.

5 Resolution: a coercive MAP objective

The Pólya–Poisson asymptote is a property of the GLD likelihood considered on Θ° alone. Adding a weakly informative log-Normal prior on the rates removes the pathology by making the objective coercive on log-rate coordinates.

Definition 10 (Weakly informative log-Normal prior). For each non-root node v and the root gain, define independently

$$\log \kappa_v \sim \mathcal{N}(m_\kappa, \sigma^2), \quad \log \lambda_v \sim \mathcal{N}(m_\lambda, \sigma^2), \quad \log t_v \sim \mathcal{N}(m_t, \sigma^2), \quad (14)$$

with default centres $m_\kappa = \log 0.1$, $m_\lambda = \log 0.5$, $m_t = \log 1$ (biologically reasonable defaults), and $\sigma = 1$.

The MAP objective is

$$\mathcal{L}^{\text{MAP}}(\Theta) := \mathcal{L}(\Theta) + \log p_\pi(\Theta) = \mathcal{L}(\Theta) - \frac{1}{2\sigma^2} \sum_{v \neq R} [(\log \kappa_v - m_\kappa)^2 + (\log \lambda_v - m_\lambda)^2 + (\log t_v - m_t)^2] + \text{const.} \quad (15)$$

Theorem 11 (Coercivity of the MAP objective). *For any $\sigma < \infty$, the MAP objective (15) is coercive on Θ° in log-rate coordinates: $\mathcal{L}^{\text{MAP}}(\Theta_n) \rightarrow -\infty$ whenever any one of $|\log \kappa_{v,n}|$, $|\log \lambda_{v,n}|$, $|\log t_{v,n}|$ tends to $+\infty$. The supremum of $\mathcal{L}^{\text{MAP}}$ on Θ° is attained at an interior point.*

Proof. For each family f , $L_f(\Theta) \in (0, 1]$, so $\log L_f(\Theta) \leq 0$. Summing over the finitely many families, $\mathcal{L}(\Theta) \leq 0$ uniformly on Θ° , hence bounded above.

The prior term in (15) is the quadratic form $-\frac{1}{2\sigma^2} \|\log \Theta - m\|_2^2 + \text{const}$ with full-rank precision $1/\sigma^2 \cdot I$, so it tends to $-\infty$ at quadratic rate as any one of $|\log \kappa_v|$, $|\log \lambda_v|$, $|\log t_v|$ goes to infinity. A bounded likelihood plus a quadratic prior that tends to $-\infty$ gives $\mathcal{L}^{\text{MAP}}(\Theta_n) \rightarrow -\infty$, which is coercivity.

Coercivity together with continuity of $\mathcal{L}^{\text{MAP}}$ (the likelihood is smooth on Θ° from the survival map; the prior is a smooth quadratic) implies every super-level set $\{\Theta : \mathcal{L}^{\text{MAP}}(\Theta) \geq c\}$, for c inside the attained range, is non-empty, bounded by coercivity, hence compact (in the log-rate coordinates) on which a continuous function attains its supremum. $\square$

Remark 12 (Calibration of $\sigma = 1$). With $\sigma = 1$ the 99% prior credible interval for each rate is $[m \cdot e^{-2.58}, m \cdot e^{2.58}]$, i.e. roughly two orders of magnitude wide. The quadratic prior cost at $\log \lambda_v = \log(5.6 \times 10^9) - m_\lambda \approx 22.5$ is $22.5^2/2 \approx 253$ nat per such node, comfortably outweighing the $\sim 20,000$ -nat asymptote gain at a single node when counted across the full prior; in practice the MAP optimum lands at $\max_v \lambda_v \approx 43$, $\max_v \kappa_v \approx 0.12$ — biologically interpretable rates, far from any boundary.

6 Numerical evidence: arc269

Dataset. The 269-leaf archaeal phylogeny of Cs26 (`tree269-baker-alti-gtdb`; $F = 90,243$ families, $N = 537$ nodes, filter $\Omega_{\min} = 1$) is the largest dataset bundled with the paper. We report fits using the consolidated driver `validation/ml.py` with clip $B = 50$ (rates in $[2 \times 10^{-22}, 5 \times 10^{21}]$).

Multiple basins, all on the asymptote. Two distinct ML optimiser trajectories are observed.

- **Simple-ML basin** (single warm-restart BFGS from default init): $\mathcal{L} = -1,129,653$, $\bar{\xi}_R = 8,664$ root copies, 3,533 root families, copies/family 2.45. Rate distribution: $\max \kappa_v = 1,540$, $\max \lambda_v = 7.6 \times 10^9$.
- **Polish basin** (multistart + lognormal polish at $\sigma = 0.10$): $\mathcal{L} = -1,109,266$ (+20,387 nat over the simple-ML basin), $\bar{\xi}_R = 8,963$ root copies, 3,608 root families, copies/family 2.48. Rate distribution: $\max \kappa_v = 1,763$, $\max \lambda_v = 5.63 \times 10^9$ (a single internal node carries the entire boundary load; the second-largest λ_v in the rate vector is 124, eight orders of magnitude below).

Both basins are sequences along the asymptote of Remark 7 at one node v_* ; the polish basin has simply climbed further up the same asymptote (cf. Proposition 8 with smaller ϵ).

Independent confirmation from the focal subsets. The same phenomenon is visible on the smaller subset clades (D80, P75, E114, ED194), but softer: Cs26’s published ML rates report $\max \kappa_v \approx 10^{14}$ on these subsets (Java GUI inspection of the bundled `.countxml.gz`). At the subset scale this drives only a $\sim 20\%$ spread in $\bar{\xi}_R$ between alternative boundary climbs; at the arc269 scale (with $\sim 20\times$ more families and a larger tree), the same mechanism inflates $\bar{\xi}_R$ by 2–3 $\times$.

MAP $\sigma = 1$ fits land in the interior. On the four focal subsets, MAP with $\sigma = 1$ lands within ± 300 nat of the published Cs26 ML values and within $\pm 20\%$ on $\bar{\xi}_R$, so the prior does not distort the non-boundary-pathological subsets. On arc269, the MAP from default uniform initialisation converges to raw $\mathcal{L} = -1,117,480$ ($-\log P_{\text{MAP}} = 1,120,615$ including the $-3,135$ -nat prior cost), with $\bar{\xi}_R = 5,906$ root copies, 3,108 root families, copies/family 1.90, $\max \lambda_v = 43$, $\max \kappa_v = 0.12$ (interior, far from any boundary). A 15-start lognormal-perturbation bootstrap discovers an even deeper basin at $-\log P = 1,098,463$, raw $\mathcal{L} = -1,096,512$, $\sim 22,000$ nat below the cold-start basin — showing that the MAP objective has multiple local maxima even at $\sigma = 1$ and that multistart is required to locate the global mode.

Method	$\mathcal{L}$	$\bar{\xi}_R$	root fam.	cp/fam	$\max \lambda_v$
Cold-start ML ($B = 50$)	$-1,129,653$	8,664	3,533	2.45	7.6×10^9
Multistart-polish ML ($B = 50$)	$-1,109,266$	8,963	3,608	2.48	5.6×10^9
MAP $\sigma = 1$ (cold start)	$-1,117,480^\dagger$	5,906	3,108	1.90	43
MAP $\sigma = 1$ (15-start global)	$-1,096,512^\dagger$	5,426	3,070	1.77	28

[†] pure log-likelihood at the MAP estimate, not the regularised objective.

7 Biological consequences and method-selection guidance

1. **ML alone is not enough for ancestral reconstruction at depth.** The two ML basins on arc269 differ only in how far along the asymptote the optimiser climbed; they do not represent

distinct biological inferences. The biological quantity of interest, the root genome size $\bar{\xi}_R$, is sensitive to the (essentially arbitrary) stopping criterion of the optimiser.

2. **The MAP $\sigma = 1$ 15-start global estimate** (root copies 5,426, root families 3,070, copies/family 1.77) is our recommended LACA quote. The cold-start MAP basin is itself an interior local optimum $\sim 22,000$ nat above the global — showing that even with the prior, multistart is required.
3. **Cs26’s published focal-subset fits are also on the asymptote.** With $\max \kappa_v \approx 10^{14}$ on D80–ED194 (log-rate ~ 32 , well below the $B = 50$ clip but well above any biologically interpretable range), every published reconstruction is technically a boundary artefact; the small magnitude on the 4-subset fits keeps the practical impact below $\pm 20\%$ in $\bar{\xi}_R$, but the underlying behaviour is the same as on arc269.
4. **The boundary issue is a model feature, not a data feature.** Any GLD reconstruction at depth without modelling horizontal gene transfer (HGT) will exhibit the same asymptote — the optimiser is asking for a Poisson immigration source, and without HGT in the model the only place to put one is at a Pólya boundary.
5. **ALE-style models including HGT may shift LACA estimates downward**, since families currently absorbed into the “inflated ancestor + asymptote” will be explained by horizontal transfer events. The MAP $\sigma = 1$ quote here is the cleanest GLD-only baseline against which to measure that shift.

A The transition-kernel limit at $\tilde{p}_v = 0$

For completeness we record the form of T_v^* (Proposition 3) when in addition $\tilde{p}_v^* = 0$ (parent-survival collapses): only the $s = 0$ term of the binomial survives, giving

$$T_v^*(m \mid n) = \text{Pois}(m \mid \Lambda_v), \quad (16)$$

independent of n . The parent count is forgotten, so the subtree below v contributes a constant per-family Poisson factor. This is the degenerate limit at which the inferred ancestor becomes literally a Poisson immigration source.

B Why a finite log-rate clip alone is a less elegant fix

A finite log-rate clip $B < \infty$ is equivalent to a uniform improper prior on $[m_a - B, m_a + B]$ in log-rate space (and 0 outside). This prior is non-smooth (a hard wall at $|\log \cdot| = B$), so the clipped MLE can sit on the boundary of the clip set itself with $\lambda_v = e^B$. With $B = 50$ in our numerics the clip is rarely attained (the optimiser stalls at $\log \lambda_{v_*} \approx 22$ due to line-search precision), so the clipped problem behaves as unbounded and inherits all the asymptote pathology characterised in Theorem 5–Proposition 8.

The MAP prior (14) achieves the same boundedness through a *smooth* quadratic cost. Both objectives are coercive, but only the smooth one yields a unique optimum that is robust to optimiser choice and stopping criterion.